INCH-POUND

MIL-DTL-46594C(AR)
11 September 2011
SUPERSEDING
MIL-C-46594B(AR)
w/AMENDMENT 2
14 JUNE 1991

DETAIL SPECIFICATION

CHARGE, DEMOLITION, SHAPED, 40 LB.
M3A1
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by U.S. Army Armament Research, Development and Engineering Center (ARDEC) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for one type of demolition kit designated as Charge, Demolition, Shaped, 40 lb., M3A1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those specified in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander ARDEC, ATTN: RDAR-QES-E, Picatinny Arsenal, New Jersey 07806-5000 or emailed to aredecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 1375
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DEPARTMENT OF DEFENSE STANDARDS AND SPECIFICATIONS

MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	-	DOD Preferred Method for Acceptance of Product
MIL-DTL-12560	-	Armor Plate, Steel, Wrought, Homogeneous (for use in combat-vehicles and for ammunition testing)

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue Bldg. 4D, Philadelphia, PA 19111-5-94.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2).

ARDEC Drawings

7549165	-	Box, Packing, Ammunition, for Charge, Demolition, Shaped, 40 lb., M3A1
7549166	-	Support, Packing, Ammunition, for Charge, Shaped, 40 lb., M3A1
8858390	-	Charge, Demolition, Shaped, 40 lb., M3A1
8858382	-	Charge, Demo, Shaped, 40 lb., M3A1, Metal Parts Assembly
8858393	-	Booster

(Copies of these drawings are available from US Army ARDEC, AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000, or online from Drawing-Request@us.army.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein.

ASTM INTERNATIONAL

ASTM E1742/E1742M	-	Standard Practice for Radiographic Examination

(Copies of ASTM standards may be ordered online at http://www.astm.org/ or from the ASTM International, 100 Barr Harbor Drive, Conshohocken, PA 19428-2959.)

NATIONAL AEROSPACE STANDARDS

NAS 410	-	NAS Certification and Qualification of Nondestructive Test Personnel

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(Copies of NAS documents may be ordered at http://aero-defense.ihs.com/collections/abstracts/nas410.htm or from the Aerospace Industries Association of America, 1000 Wilson Blvd., Arlington, VA 22209.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supercedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified, a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.1 Radiographic examination. Radiographic examination, in accordance with 4.3.3.3 and 6.5, shall be conducted on First Article samples that are to be subjected to the Functioning Test, prior to the execution of the Functioning Test (see Table II). Radiographic examination data shall not be used to accept/reject this system.

3.2 Material. Materials, parts and assemblies shall comply with all requirements specified on the applicable drawings and referenced specifications.

3.3 Assemblies. The assemblies shall comply with all requirements specified on applicable drawings and with all requirements specified in the applicable specifications.

3.4 Functioning Test. The loaded assembly shall function satisfactorily as evidenced by high-order detonation of the charge and penetration of a target of Rolled Homogeneous Armor (RHA) in accordance with MIL-A-12560, Class 1 or 2. The RHA target shall be of a size not less than four inches total thickness and surface area not less than 144 square inches, with any number of plates to make up that size. The penetration hole in the plate shall exhibit no evidence of double- or multi-jets. This test shall be performed as specified in 4.3.3.4.

3.5 Workmanship. Loading and assembling shall be accomplished in a thorough workmanlike manner. All parts and assemblies shall be clean and free of explosive on exterior, chips, cracks, crazes, dirt, grease, rust, corrosion products, and other defects and foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part or assembly nor shall the cleaning agent contaminate them. Exterior surface coatings shall be continuous. All required markings shall be neat and sharply defined. Required packing shall be dry.

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4. VERIFICATION

Table I. REQUIREMENTS/VERIFICATION CROSS REFERENCE MATRIX

Method of Verification	Classes of Verification
1- Analysis 2- Demonstration (end item test) 3- Examination 4- Test	A - First Article B - Conformance

Section 3 Requirements	Description	Verification Method				Verification Class		Section 4
		1	2	3	4	A	B	
3.1	First Article	X		X	X	X		4.2
3.1.1	X-ray examination	X		X		X		4.3.3.3
3.2	Material	X		X	X	X	X	4.3
3.3	Assemblies			X	X	X	X	4.3
3.4	Functioning Test		X			X	X	4.3
3.5	Workmanship	X		X	X	X	X	4.3

4.1 Classification of Verification. The verification requirements specified herein are classified as follows:

a. First Article Inspection (see 4.2)

b. Conformance Inspection (see 4.3)

4.2 First article inspection.

4.2.1 Submission. When specified, a first article sample shall be submitted for evaluation in accordance with the provisions of 4.2.2 and Table II.

4.2.2 Inspections to be performed. As determined by the Government, the first article samples may be subjected to any or all of the examinations and tests specified in the specification and inspected for compliance with any or all of the requirements of this specification and applicable drawings, including all unlisted characteristics.

4.2.2.1 Rejection. The first article sample shall be rejected if any of the inspections/tests fail to meet specification or drawing requirements. The Government reserves the right to terminate inspection or test upon failure of any sample to comply with any requirement.

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Table II. FIRST ARTICLE INSPECTION

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Charge, Demolition, Shaped, 40 lb. M3A1		SHEET 1 OF 1		DRAWING NUMBER (See below) NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
	Radiographic examination	5*	100%	3.1.1	4.3.3.3
	Functioning Test	5*	100%	3.4	4.3.3.4
	Parts Assemblies				
	Examination of defects (where applicable)	25	100%	3.3	4.3
	Booster				
	Examination of defects and Specific Gravity (Drawing (Dwg.) 8858393)	5	100%	3.2	4.3
	Loading Assemblies				
	Examination of defects (where applicable) and Weight of Charge (Dwg. 8858390)	25	100%	3.3	4.3

NOTE*: Radiographic examination shall be conducted on the First Article samples that are to be subjected to the Functioning Test, prior to the execution of the Functioning Test. Radiographic examination data will not be used to accept/reject this system.

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4.3 Conformance Inspection.

4.3.1 Inspection lot formation. The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-1916. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded charge assemblies shall contain:

a. Parts from not more than one lot interfix number from one manufacturer.

b. Composition B of the same grade from one lot interfix number from one manufacturer which contains TNT from not more than one lot interfix number from one manufacturer.

c. Composition A-3 of the same grade from one lot interfix number from one manufacturer.

4.3.2 Examination. Unless otherwise specified in the Conformance Inspection, sampling plans for major and minor defects shall be in accordance with MIL-STD-1916, Verification Level IV for Major defects and Verification Level II for Minor defects.

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Conformance Inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.1	TITLE Booster	SHEET 1 OF 1	DRAWING NUMBER 8858393 NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major					
101	Concentricity of small cavity to outside diameter		IV	3.2	Gage
102	Diameter of small cavity, minimum (min.)		IV	3.2	Gage
103	Depth of Large cavity, min.		IV	3.2	Gage
104	Specific gravity	5		3.2	4.3.3.2
Minor					
201	Depth of small cavity, min.		II	3.2	Gage
202	Diameter of large cavity, min.		II	3.2	Gage
203	Total length, maximum (max.)		II	3.2	Gage
204	Evidence of poor workmanship		II	3.5	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.2	TITLE Assembly, Prior to Loading	SHEET 1 OF 1	DRAWING NUMBER 8858390 NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None Defined				
Major					
101	Interior coating with bare spot		IV	3.3	Visual
102	Foreign matter in interior		IV	3.3	Visual
Minor					
201	Evidence of poor workmanship		II	3.5	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.3	TITLE Assembly, prior to closing		SHEET 1 OF 1		DRAWING NUMBER 8858390 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None Defined				
Major					
101	Charge improper		IV	3.3	Visual
102	Booster missing (M3A1 only)		IV	3.3	Visual
103	Pad or spacer missing (M3A1 only)		IV	3.3	Visual
104	Container damaged		IV	3.3	Visual
Minor					
201	Depth to charge		II	3.3	Gage
202	Evidence of poor workmanship		II	3.5	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.4	TITLE Assembly		SHEET 1 OF 1		DRAWING NUMBER 8858390 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major					
101	Gage does not fully enter detonator well		IV	3.2	Gage
102	Closing cap assembly not tight or adhesive missing		IV	3.2	Visual/Manual
103	Protective tape missing		IV	3.2	Visual
Minor					
201	Evidence of poor workmanship		II	3.5	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.5	TITLE Loading and marking assembly		SHEET 1 OF 1		DRAWING NUMBER 8858390 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major					
101	Component missing		IV	3.3	Visual
102	Strap slide improperly assembled		IV	3.3	Visual
103	Functioning Test	5		3.4	4.3.3.5
104	Weight of charge	25		3.3	4.3.3.3
Minor					
201	Total length, max.		II	3.3	Gage
202	Width at stand-off legs, max		II	3.3	Gage
203	Color coding improper		II	3.3	Visual
204	Marker missing, misleading or unidentifiable		II	3.5	Visual
205	Evidence of poor workmanship				Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.6	TITLE Box, packing, prior to closing		SHEET 1 OF 1		DRAWING NUMBER 7549165 or 7549166 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major					
101	Support missing		IV	3.2	Visual
102	Packing component missing		IV	3.2	Visual/Manual
Minor	None defined				

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Conformance Inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.3.2.7	TITLE Box, packing, sealed		SHEET 1 OF 1		DRAWING NUMBER 7549165 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined				
Major					
101	Contents of box exposed		IV	3.2	Visual
102	Board broken		IV	3.2	Visual
103	FSC number missing or misleading		IV	3.2	Visual
104	Hardware missing, broken or loose		IV	3.2	Visual
105	Strapping missing, broken or loose		IV	3.2	Visual
106	Contents loose when box is inverted		IV	3.2	Visual
Minor					
201	Marker missing, misleading or unidentifiable		II	3.2	Visual
202	Hardware improperly engaged		II	3.2	Visual/Manual
203	Strapping improperly engaged		II	3.2	Visual/Manual
204	Car seal missing, unsealed or improperly positioned		II	3.2	Visual/Manual
205	Handle missing or insecure		II	3.2	Visual/Manual

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4.3.3 Test Methods and Procedures.

4.3.3.1 Specific gravity of booster. A sample of five (5) boosters shall be selected from each lot for this test. If any booster fails to meet the requirement specified, the lot shall be rejected. The booster shall be weighed on a balance accurate to 0.1 grams and the weight recorded as dry weight. The booster shall then be weighed while completely submerged in water, taking care to remove all air bubbles inside the detonator well and remaining on the surfaces of the booster. The temperature of the water shall be not less than 66 degrees Fahrenheit (^{o}F) and not more than 70 ^{o}F. This weight shall be recorded as weight under water. (When submerged, the position of the booster shall be such that no air is entrapped.) Specific gravity shall be calculated as follows:

$$\textit{Specific gravity} = \frac{\textit{Weight of booster dry}}{\textit{(Weight of booster dry) minus (Weight of booster immersed in water)}} \times \textit{Specific gravity of } H_2O \textit{ at noted temperature (see Table III).}$$

Table III. Specific gravity of water within temperature range

Temp °F	Temp °C	Specific Gravity of H_2O
66	18.89	0.9984
67	19.44	0.9983
68	20.00	0.9982
69	20.56	0.9981
70	21.11	0.9980

4.3.3.2 Weight of charge. A sample of 25 shaped charges shall be selected from each lot and weighed on a scale capable of determining weights within an accuracy of two (2) ounces. The average weight of 30 metal parts assemblies shall be subtracted from each individual weight to determine compliance with the requirement. Any unit failing to meet the weight requirements shall be classed defective.

4.3.3.3 Radiographic examination. Radiographic examination shall be conducted on the First Article samples that are to be subjected to the Functioning Test (see 4.3.2.5, Major 106), prior to the execution of the Functioning Test. The First Article sample of five (5) test units shall be subjected to radiographic examination for detection of voids, cracks, and foreign material. Examinations of the radiograph shall be made to determine excessive cavitation, separation, and cracking of the explosive and presence of any foreign material. Radiographic equipment, operations and procedures shall be qualified in accordance with ASTM E 1742 (see 6.5). Radiographic personnel shall be qualified in accordance with NAS 410. Complete results shall be available to the Government (see 6.5).

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4.3.3.4 Functioning test. Five (5) test units shall be selected from each lot. The loaded assembly shall be assembled for firing and positioned on the steel plate. A M6 or M7 blasting cap or equivalent military initiation device shall be secured in the capwell and the unit initiated. Observation shall be made for the presence of undetonated charge and for evidence of double or multi-jets. Presence of either shall class a unit defective.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2f). When packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The 40 lb. Shaped Demolition Charge is primarily employed for burring holes in earth, metal, concrete, and paved or unpaved roads

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article samples.

c. Requirements for Ammunition Lot Numbering, if applicable.

d. Requirement for acceptance inspection equipment (AIE) designs.

e. Certificate of Conformance for each lot or shipment of product.

f. Preservation, packaging, and marking.

1. Level A. The M3A1 Loaded Assembly is packed and marked in accordance with Dwg. 7549165.

6.3 Acceptance inspection equipment. Provisions concerning the AIE and test facilities that will be used to verify the requirements of this specification should be specified in the contract.

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6.4 Distribution of ammunition data cards. Distribution of data cards should include the following: Commander RDECOM-ARDEC, U.S. Army, ATTN: AMSRD-AAR-QEM, Picatinny, NJ 07806-5000.

6.5 Radiographic examination. Complete results of the radiographic examination should be available to the Government through the Contracting Officer. Distribution shall include: U.S. Army RDECOM-ARDEC, AMSRD-AAR-AEM, Picatinny, NJ 07806-5000.

6.6 Certification. Unless otherwise specified by the Contracting Officer, certification should be submitted by the contractor in accordance with the contract. Distribution should include: U.S. Army RDECOM-ARDEC, AMSRD-AAR-AEM, Picatinny, NJ 07806-5000.

6.7 Subject term (key word) listing.

Detonation
Kit
Munitions

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army-AR

Preparing Activity:
Army-AR
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